

Numerical Exercise #5

Jordi Mitjà

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1 Question #1 Analytical solution

U235	2.3100×10^{20}
U238	7.4700×10^{21}
Pu239	0.0000×10^0
X	0.0000×10^0
Y	0.0000×10^0

Table 1: Initial atomic concentrations in at/cm^{-3}

Governing differential equations for each isotope (U235, U238, Pu239, X and Y):

$$\begin{aligned}
 \frac{dN_{U238}}{dt} &= -\Phi \sigma_{a,238} N_{U238}, \\
 \frac{dN_{U235}}{dt} &= -\Phi \sigma_{a,235} N_{U235}, \\
 \frac{dN_{Pu239}}{dt} &= \Phi \sigma_{\gamma,238} N_{U238} - \Phi \sigma_{a,Pu} N_{Pu239}, \\
 \frac{dN_X}{dt} &= y \Phi \sigma_{f,235} N_{U235} - \lambda_X N_X, \\
 \frac{dN_Y}{dt} &= y \Phi \sigma_{f,235} N_{U235} - \lambda_Y N_Y.
 \end{aligned} \tag{1}$$

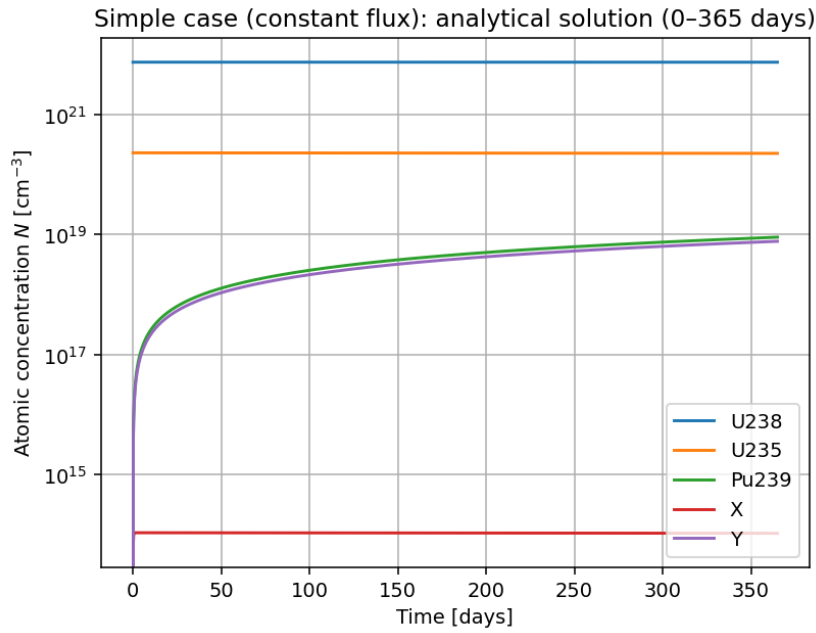


Figure 1: Evolution of the various isotopes concentrations in a semilog-Y scale between 0 and 365 days

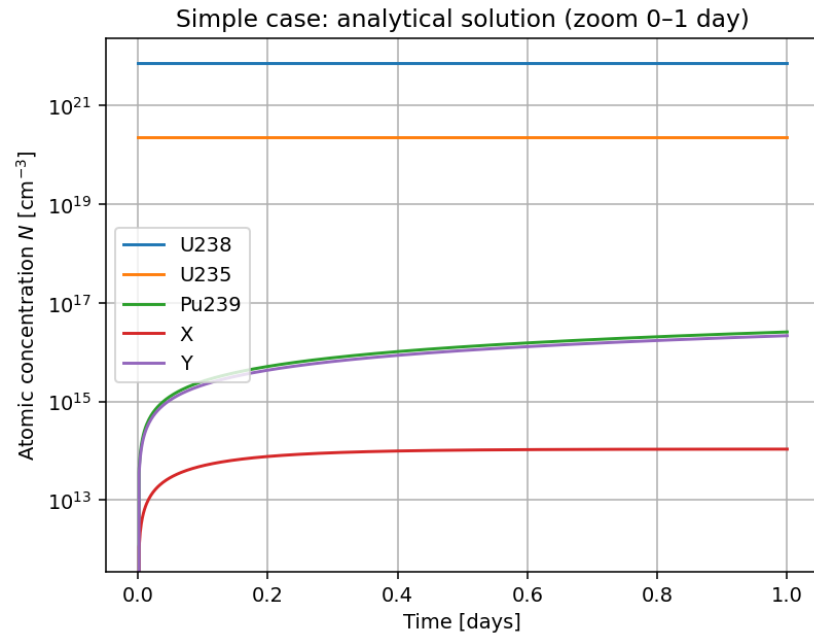


Figure 2: Evolution of the various isotopes concentrations in a semilog-Y scale between 0 and 1 day

2 Question #2 Numerical solution

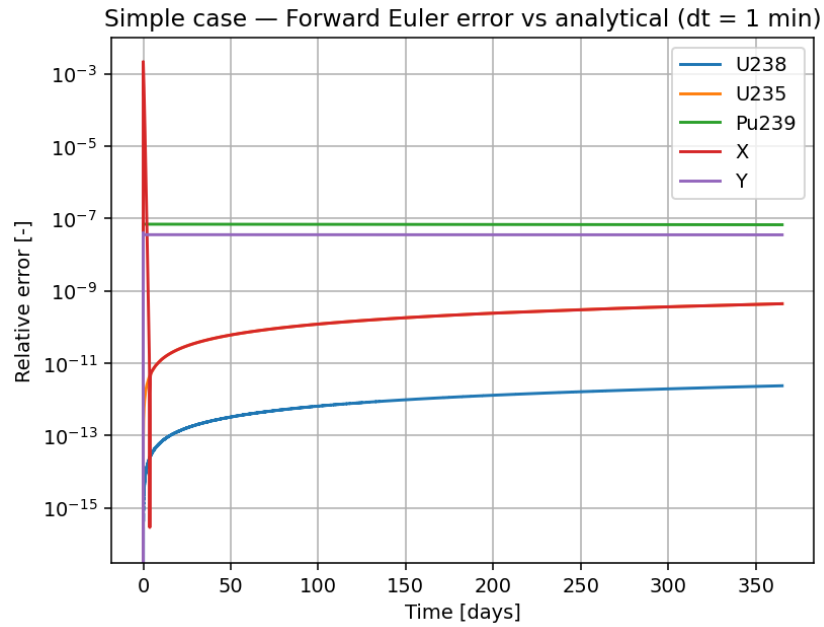


Figure 3: Evolution of the error with respect to the analytical solution; for the various isotopes concentrations in a semilog-Y scale between 0 and 365 days with the Forward Euler method and a time-step of 1min

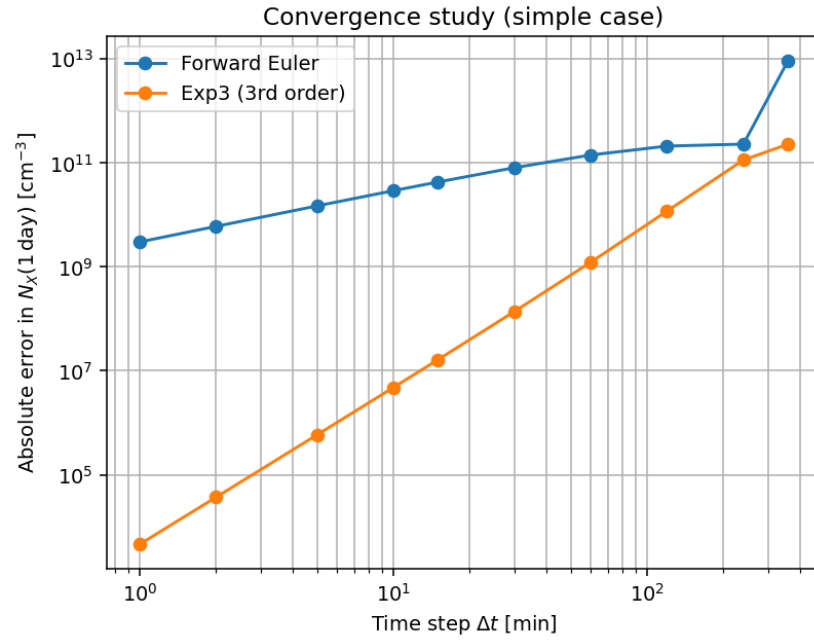


Figure 4: Illustration of the convergence rate of Euler and Matrix exponential methods, varying the time step between 1min and 6h. The quantity of interest is the concentration of X after 1 day.

3 Question #3 Advanced Problem

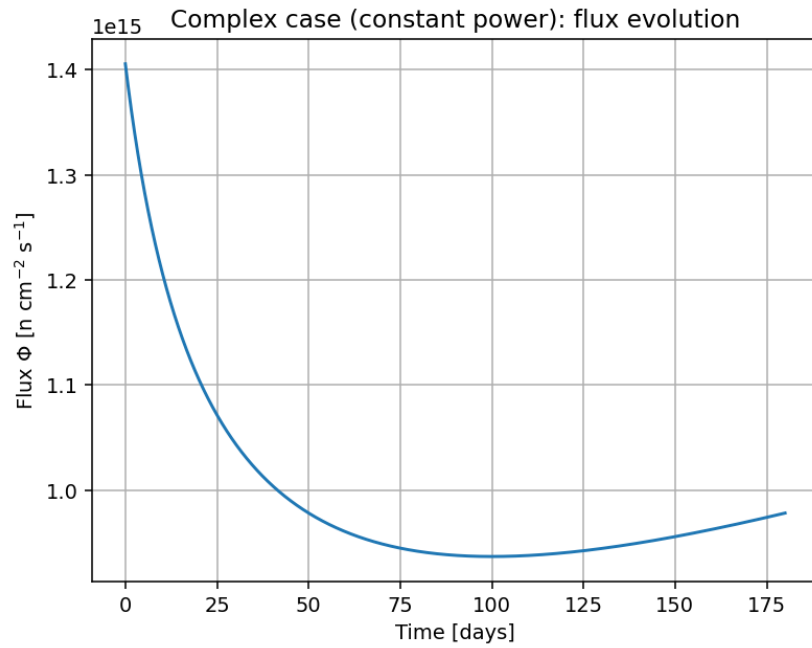


Figure 5: Evolution of the flux level ($\phi [cm^{-2} s^{-1}]$) as a function of time.

Explain the observed trends in Figure 5.

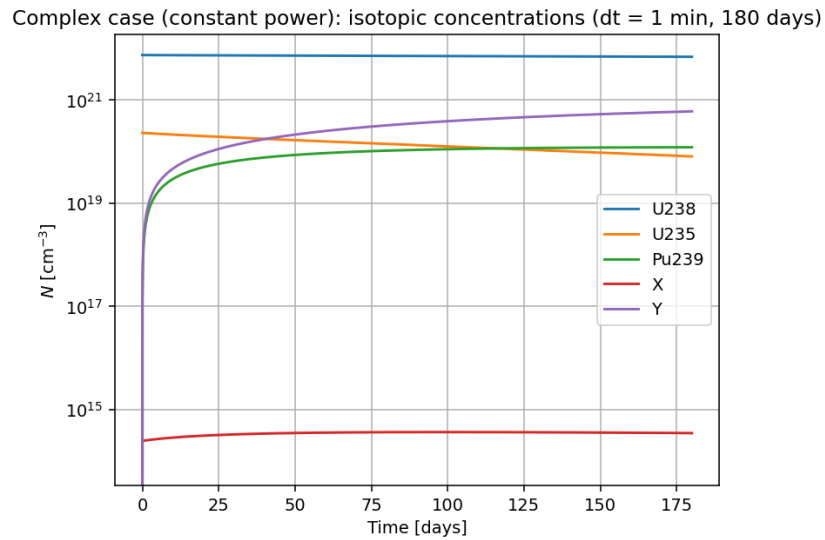


Figure 6: Evolution of the various isotopes concentrations in a semilog-Y scale between 0 and 180 days.